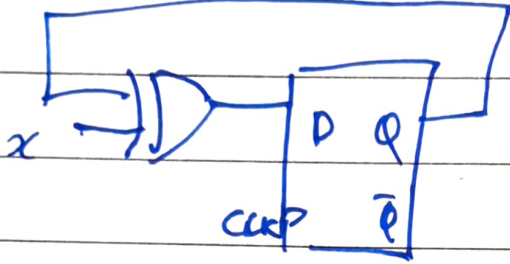


Day 4 Assignment

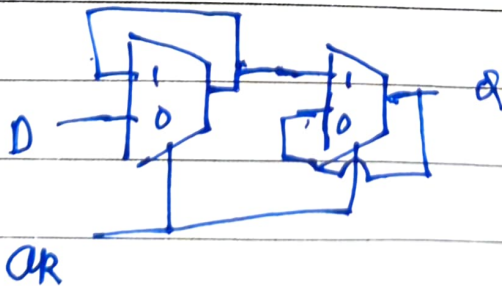
Q1



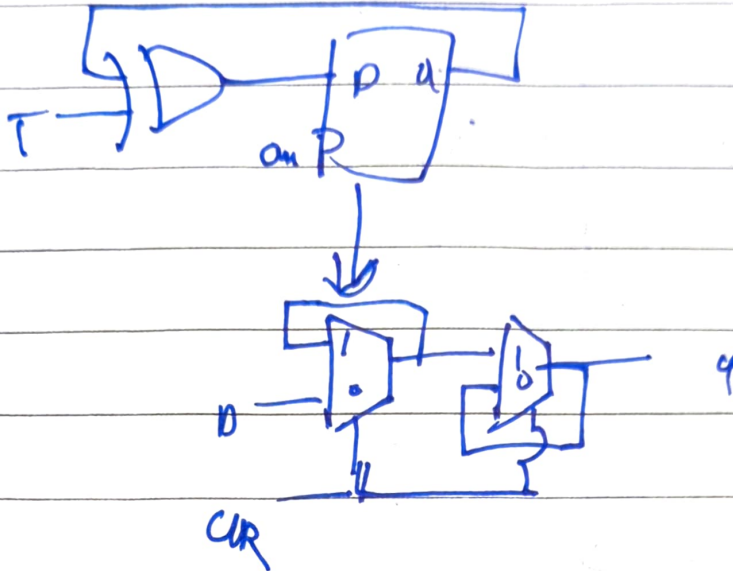
x	Q_{prev}	Q_{next}
0	0	0
0	1	1
1	0	1
1	1	0

it behaves like a T flipflop

Q2 $= \bar{S} I_0 + S I_1$ DFF using 2:1

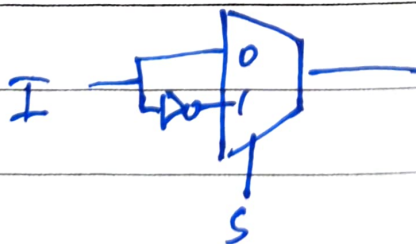
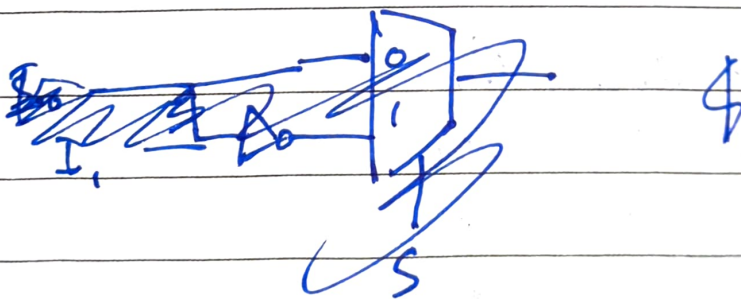


Q3, T flipflop using mux



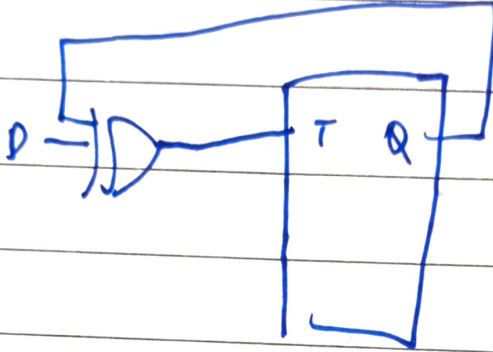
XOR $\rightarrow$

$$\overline{S}I_0 + S\overline{I}_1 \Rightarrow T \text{ mode}$$

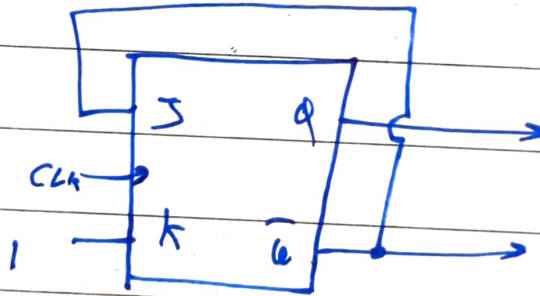


Now cascading both we get TFF

Q4 T to D



Q5



$\bar{Q}_p$

J	K	$\bar{Q}_p$	Q_n	$\bar{Q}_n$
1	1	0	0	1
0	1	1	0	1
1	1	0	1	0
0	1	1	0	1
1	1	0	1	0
			0	1